



Tutorial S2A: Programming Elements

Section A (Discussion Questions)

1. **[test data]** An array contains eight positive integers. An algorithm is designed to locate the first incidence of a particular integer in the array. Test data is required to test the algorithm as fully as possible. The table shows 2 suitable examples of test conditions and test data:

Condition being tested	Array data	Search integer
Search integer not present	5 8 5 9 2 3 9 4	7
Array contains an incorrect amount of integers	2 3 5 6 4 8 1	9

Copy the above table headings and write down two more conditions which should be tested, providing suitable test data in each case.

2. **[trace table]** An algorithm is given below:

```
DECLARE X: INTEGER
DECLARE Y: INTEGER
DECLARE Z: INTEGER
DECLARE I: INTEGER
START MAIN PROGRAM
  INPUT X
  INPUT Y
  Z ← 1
  FOR I ← 1 to Y
    Z ← Z * X
  ENDFOR
  OUTPUT Z
END MAIN PROGRAM
```

- (a) Complete the table below to show how each variable changes when the algorithm is performed on the test data given.

Test data: $x = 2$ and $y = 4$

I	X	Y	Z

- (b) Briefly describe the purpose of this algorithm.
3. **[recursion]** Define $\sum_{r=1}^N r = 1 + 2 + 3 + 4 + \dots + N$ as a recursive function as follows:
SUM_1toN(N: INTEGER) RETURNS INTEGER
- (a) Write the pseudocode for the function.
- (b) Draw a trace tree diagram for the recursive function call for SUM_1toN(4).
- (c) What is/are the total number of time(s) SUM_1toN(4) is called in the main program?
- (d) What is/are the total number of time(s) SUM_1toN(4) calls itself?

4. **[program errors]** The program given below is supposed to calculate and display the total payable (inclusive of the 7% GST) for purchasing a number of items from DAISO where every item is sold at \$2.00 per piece. The program code is given as:

```
1  gst = 7
2  unitprice = 2.0
3  print(Welcome to DAISO)
4  quantity = input("Enter the number of items: ")
5  cost = quantity * unitprice
6  gstcost = cost * gst
7  total = cost + gstcost
8  print("Total payable: ", total)
```

Identify three errors in the program code and complete the following table.

Line number	Explanation of error	Correction of error

5. **[data validation]** Validation is a way of checking data in a database. Name the most appropriate validation check that matches each description below.

- (a) Ensure that data entered into a field is exactly 10 characters
- (b) Ensure that a number lies between 10 and 100
- (c) Ensure that the data entered is numeric
- (d) Ensure that the date is entered as DD/MM/YYYY

6. **[data validation]** An airline keeps a database of all the flights it operates. A short extract is shown here.

Flight_number	Departure_Airport_Code	Number_of_passengers	Destination_Airport_Code	Ticket_prices
EK236	JNB	135	LHR	\$1200
EK080	DXB	256	DEL	\$500
EK029	LHR	375	DXB	\$650

Explain what is meant by the following validation checks using the extract above.

- (a) Range check (b) Format check (c) Length check

7. **[test data]** A program gets the user to input an odd integer and output a pyramid with width specified by the integer. For example, if user input is 9, the pyramid would look like this:

Enter width (an odd number): 9

Explain how you would select test data for this program.

8. [trace table] A segment of an algorithm is as shown.

```

S ← 0
E ← 7
F ← False
P ← -1

INPUT SV

REPEAT

    M ← (S + E) DIV 2

    IF SV = Array[M] THEN
        F ← True
        P ← M
    ENDIF

    IF SV > Array[M] THEN
        S ← M + 1
    ENDIF

    IF SV < Array[M] THEN
        E ← M - 1
    ENDIF

UNTIL (F = True) OR (E < S)

```

Test Data

Array	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]
	13	27	33	49	55	67	75	79

SV = 75

DIV is integer division (floor division).

It gives the quotient that is obtained from division.

e.g. $7 \text{ DIV } 2 = 3$

Complete the table below to show how each variable changes when the algorithm is performed on the test data given above.

S	E	M	Array[M]	F	P
0	7	3	49	False	-1
4	7				

9. [recursion/trace table]

(a) Explain briefly the main features of a recursive procedure from the programmer's point of view. Explain what is required from the computer system in order to enable recursion to be used.

(b) The following recursive subroutine carries out a list operation.

```

01 FUNCTION listProcess(numList)
02   IF LENGTH(numList) > 0 THEN
03     Remove first element of numlist and store in first
04     listProcess(numList)
05     Append first to end of numList
06   ENDIF
07   RETURN numList
08 ENDFUNCTION

```

(i) Complete the following trace table if the list numbers is defined in the main program as `numbers = [3, 5, 10, 2]` and the subroutine is called with the statement `new = listProcess(numbers)`

	numList					
LENGTH(numList)	0	1	2	3	first	new
4	3	5	10	2	3	

(ii) Explain what the subroutine does.

10. [pseudocode/recursion] The words COW, BEEF, FORTY have all their letters written in alphabetical order. Here is an algorithm for a function which checks whether all the letters in a word are in alphabetical order.

```

01 FUNCTION IsInOrder (Word)
02     IF LENGTH (Word) = 1 THEN
03         RETURN TRUE
04     ELSE
05         FirstChar = First character in Word
06         RestOfWord = All characters in Word except the first
07         IF FirstChar > RestOfWord THEN
08             RETURN FALSE
09         ELSE
10             RETURN IsInOrder (RestOfWord)
11         ENDIF
12     ENDIF
13 END FUNCTION

```

- (a) Describe what is meant by a parameter and give an example from the algorithm above.
- (b) Explain the difference between the uses of the = sign in line 02 and in line 05, stating the type of operation being carried out.
- (c) Line 07 compares the first character of the word with the rest of the word. Explain why there may be a problem with the call `IsInOrder ("FoRtY")` and what can be done to avoid this problem.
- (d) State what is meant by recursion, using the algorithm above as an example.
- (e) The algorithm is tested with the call `IsInOrder ("Z")`. State the value which will be returned. State the lines of the algorithm which will be executed.
- (f) Explain what happens when the algorithm is tested with a call `IsInOrder ("")` where the value of the argument is the empty string.
- (g) Explain what happens when the algorithm is tested with a call `IsInOrder ("APE")`. Use a diagram to show each call made, the lines of the algorithm executed and the return value of each call.

11. 9597/2014/02/06 [program errors]

A function is to be written that returns the sum of all values held in an array that are greater than a minimum value. The function will be used with arrays of varying size, but never more than a maximum of 50 000 elements.

A first attempt at writing the program code for the function is given below:

```
01 FUNCTION TotalSum(Results : ARRAY[50000] OF REAL,  
                    ArraySize : INTEGER, MinValue : REAL)  
    RETURNS REAL  
02     DECLARE Sum, Counter : INTEGER  
03     DECLARE Temp: Real  
04     Sum = 0.0  
05     FOR Counter = 1 TO ArraySize  
06         Temp = Results[Counter]  
07         IF Temp > MinValue THEN Sum = Sum * Temp  
08     ENDFOR  
09     RETURN Sum  
10 ENDFUNCTION
```

The function is included in a program specifically written to test the function. The main program outputs the value returned by the function.

A compiler was used to compile the source program.

- (a) The compiler reported an error at line 5 in the function. Identify the error and explain why it was flagged as a syntax error.
- (b) The compiler also reported an error at line 4. State the type of error reported by the compiler justifying your answer.

The errors indicated in **parts (a) and (b)** were corrected. A successful compilation produces executable code. When the code was executed, the program failed to complete and reports an error at line 7.

- (c) (i) State the type of error that occurred. Justify your answer.
(ii) The error described in part (c)(i) depends on the detection of another type of error. Name this other type of error. How should the code be changed to correct this error?

When the program finally runs without error, the test plan needs to be completed. The test plan uses data that tests different sizes of array, different array values and different minimum values.

The array TempArray is used in the main program as the array to be processed.

- (d) Each element of TempArray stores a random value between 1.0 and 10.0.
 - (i) Explain why the function call: TotalSum(TempArray, 1000, 5.0) is not an appropriate black box test¹.
 - (ii) Explain why the function call: TotalSum(TempArray, 10, 10.5) is not an appropriate white box test².
- (e) If each element of TempArray stores the value 1.0, state a function call that will be an appropriate black box test. Justify your answer.

¹ Tester uses test data, for which the results are already known, and compares the results from the program with those that are expected. Tester does not look at the source code, only look at input and output.

² Tester examines each line of code for correct logic and accuracy. May draw up a trace table to record values of variables after each instruction is executed.

Section B (Assignment Questions)

1. **[recursion]** The recursive function `ToFind` has two parameters, `x` and `y`.

```

01  FUNCTION ToFind(x: INTEGER, y: INTEGER) RETURNS INTEGER
02      IF y = 0
03      THEN
04          RETURN x
05      ELSE
06          RETURN ToFind(y, x MOD y)
07      ENDIF
08  ENDFUNCTION

```

Note: $x \text{ MOD } y$ gives the remainder when x is divided by y .

- (a) Draw a trace diagram for the recursive function call `ToFind(24, 60)`, showing for each re-entry into the `ToFind` function, the values passed to the function and the results returned.
- (b) Explain the purpose of the function `ToFind`.
- (c) Write the pseudocode for the iterative version of `ToFind`. Name it `ToFindIte`.

2. **[validation & errors]** An ISBN 10 (International Standard Book Number) is a 10-digit number that is used to identify a book. The first nine digits of the ISBN number are used to represent the Title, Publisher and Group of the book and the last digit is a check digit that is calculated from the first nine digits using the modulus 11 system. The steps are shown below:

own below:

ISBN (first 9 digits)	0	7	5	1	5	4	9	2	6	
Weight	10	9	8	7	6	5	4	3	2	
Sum of the Product	0	63	40	7	30	20	36	6	12	= 214
Remainder	214 divided by 11 = 19 remainder 5									
Check Digit	11 – 5 = 6			If the check digit is 10, it is replaced by ‘X’ If the check digit is 11, it is replaced by ‘0’						
ISBN	0 7 5 1 5 4 9 2 6 6									

- (a) Write, in pseudocode, a function `GenerateCheckDigit`, to calculate the check digit of an ISBN number. The function takes in one argument which is the 9-digit ISBN number and returns the check digit.

```
FUNCTION GenerateCheckDigit (ISBN : STRING) RETURNS CHAR
```

- (b) Explain the purpose of using a check digit. Explain two types of errors which a check digit can capture.